

$\vartheta$ : when a coordinate communicates

independent of

$q_r$ : how far the server model moves

each client  $i = 1, \dots, M$

server

all clients

1 LOCAL SGD

$$\mathbf{w}_{i,0}^r = \mathbf{w}^r$$

$$\boldsymbol{\delta}_i^r = -\eta_r \sum_{e=0}^{E_r-1} \mathbf{g}_{i,e}^r$$

local model change

2 LEAKY EVIDENCE

$$\tilde{\mathbf{z}}_i^r = \rho_r \mathbf{z}_i^r + \frac{p_i}{\eta_r} \boldsymbol{\delta}_i^r$$

retained state

new evidence



state carried  
across rounds

3 THRESHOLD AND RESET

$$m_{ij}^r = \mathbf{1}_{\{|\tilde{z}_{ij}^r| \geq \vartheta\}}$$

$$c_{ij}^r = m_{ij}^r \text{sign}(\tilde{z}_{ij}^r)$$

$$z_{ij}^{r+1} = (1 - m_{ij}^r) \tilde{z}_{ij}^r$$

fire: send  $c_{ij}^r = \pm 1$  and reset  $z_{ij}^{r+1} = 0$



sparse  
uplink  
address  $j$   
+ sign  $c_{ij}^r$



4 SERVER AGGREGATION

$$\mathbf{C}^r = \sum_{i=1}^M \mathbf{c}_i^r$$

$$\mathbf{w}^{r+1} = \mathbf{w}^r + q_r \mathbf{C}^r$$

events reinforce or cancel

one net event moves the model by  
 $q_r$

5 EXACT SYNCHRONIZATION

ordered sparse replay

OR

complete float32 checkpoint

whichever  
is smaller

$$\mathbf{w}_{i,0}^{r+1} = \mathbf{w}^{r+1}, \quad i = 1, \dots, M$$

clients exactly match the server

next round starts from  $\mathbf{w}^{r+1}$